

MOBILITY PRICING INTELLIGENCE: COST PREDICTION & PRICING DRIVER ANALYSIS

Inventory Planning, Pricing Dynamics
& Operational Performance



TOOLS USED

Python, Pandas, Numpy, Matplotlib, Plotly &
ML libraries

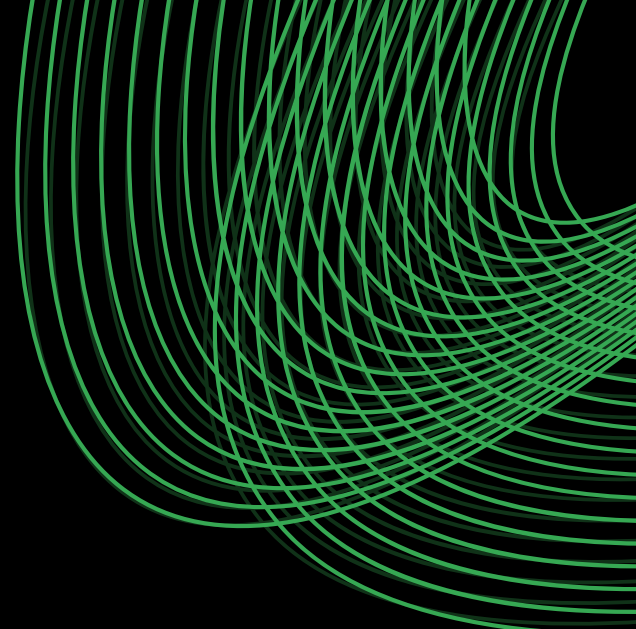
PRESENTED BY

Tejas Jadhav



TABLE OF CONTENTS

Overview & Business Objective	3
Data Understanding & EDA	8
Data Preparation & Feature Engineering	22
Model Development & Experimentation	29
Model Evaluation & Comparison	34
Explainability & Final Model	44
Business Insights & Recommendations	49
Conclusion	54



Overview & Business Objective



1.1 PROJECT OVERVIEW

Retail businesses operate in a dynamic environment where demand is influenced by factors such as historical sales, inventory, pricing, promotions, seasonality, and region.

Accurate demand estimation helps businesses maintain optimal inventory, reduce stockouts and excess stock, and improve procurement and operational planning.

The Retail Demand Intelligence project uses historical retail data and machine learning to predict demand and identify the key factors influencing it.

The project combines:

- Data understanding and cleaning
- Exploratory Data Analysis
- Feature engineering
- Machine learning regression
- Model evaluation and comparison
- Prediction-error analysis
- Model explainability
- Business-oriented insights

The objective is not only to predict demand accurately, but also to understand why the model makes its predictions & how these insights can support better decision-making.



1.2 PROBLEM STATEMENT



Retail demand is influenced by multiple interacting variables, making manual forecasting or reliance on historical averages insufficient for many situations.

The key problem addressed in this project is:

How can historical retail data be used to build a machine learning model that accurately predicts product demand while also identifying the major factors driving demand?

The project therefore treats Demand as the target variable and uses historical, operational, pricing, promotional, environmental, and temporal information as predictive features.



1.3 PROJECT OBJECTIVES

The major objectives of the project are:

- Develop a machine learning-based retail demand prediction system.
- Understand the underlying structure and characteristics of the dataset.
- Identify important patterns through exploratory data analysis.
- Create meaningful temporal, inventory, pricing, and historical features.
- Train and compare multiple regression models.
- Evaluate model performance using appropriate regression metrics.
- Investigate the largest prediction errors.
- Identify the most influential demand-driving features.
- Identify limitations and possible improvements for future development.



1.4 PROJECT SCOPE

The project follows a structured workflow covering data analysis, demand modeling, model explainability, and business interpretation to understand demand patterns, evaluate predictive performance, identify key demand drivers, and derive actionable retail insights.

01

Data Analysis

- Dataset inspection
- Data quality assessment
- Distribution analysis
- Relationship analysis

02

Demand Modeling

- Feature preparation
- Regression modeling
- Model comparison
- Model evaluation

03

Explainability

- Feature importance analysis
- Original-feature aggregation
- Investigation of large prediction errors

04

Business Interpretation

- Business Interpretation
- Inventory implications
- Pricing and promotion observations
- Seasonal and weather-related effects
- Recommendations for retail planning



Data Understanding & Exploratory Data Analysis



2.1 DATA OVERVIEW

The dataset provides a comprehensive view of retail operations across stores and products, covering inventory levels, sales activity, ordering patterns, pricing, discounts, promotions, weather conditions, seasonality, and demand. These variables provide the foundation for analyzing demand behavior, identifying operational patterns, and developing predictive models for retail demand.

Data Source: Kaggle

Dataset Size

76,000

Records

16

Original Features

Purchase Diversity

4

Regions

5

Stores

20

Products



DATA OVERVIEW

Below is a detailed description of the feature set:

Dataset Features	Type	Feature Description
Date	Date / Time	Date of the observation
Store ID	Categorical	Unique identifier of the store
Product ID	Categorical	Unique identifier of the product
Category	Categorical	Product category
Region	Categorical	Geographic region of the store
Inventory Level	Numerical (Discrete)	Inventory available at the time of observation
Units Sold	Numerical (Discrete)	Number of units sold
Units Ordered	Numerical (Discrete)	Number of units ordered
Price	Numerical (Continuous)	Selling price of the product
Discount	Numerical (Discrete)	Discount applied to the product
Weather Condition	Categorical	Weather condition during the observation
Promotion	Binary	Indicates whether a promotion was active
Competitor Pricing	Numerical (Continuous)	Price of the comparable product offered by competitors
Seasonality	Categorical	Seasonal classification of the observation
Epidemic	Binary	Indicates whether an epidemic-related condition was present
Demand	Numerical (Discrete)	Target variable representing product demand

2.2 TARGET VARIABLE

The target variable selected for prediction is:

Demand

Demand represents the quantity that the model attempts to estimate based on the available information

The target has:

104.32

Mean

100

Median

4

Minimum

430

Maximum

46.96

Standard Deviation

The distribution is concentrated around approximately 70-130 units, with a noticeable right tail extending toward higher demand values.



2.3 DATA QUALITY & PREPARATION OVERVIEW

Before modeling, the dataset was examined for:

- Data types
- Missing values
- Duplicate observations
- Categorical variables
- Numerical variables
- Temporal information
- Potential outliers
- Feature relationships

The *Date* variable was converted into usable temporal components, while categorical variables were prepared for machine learning using one-hot encoding.





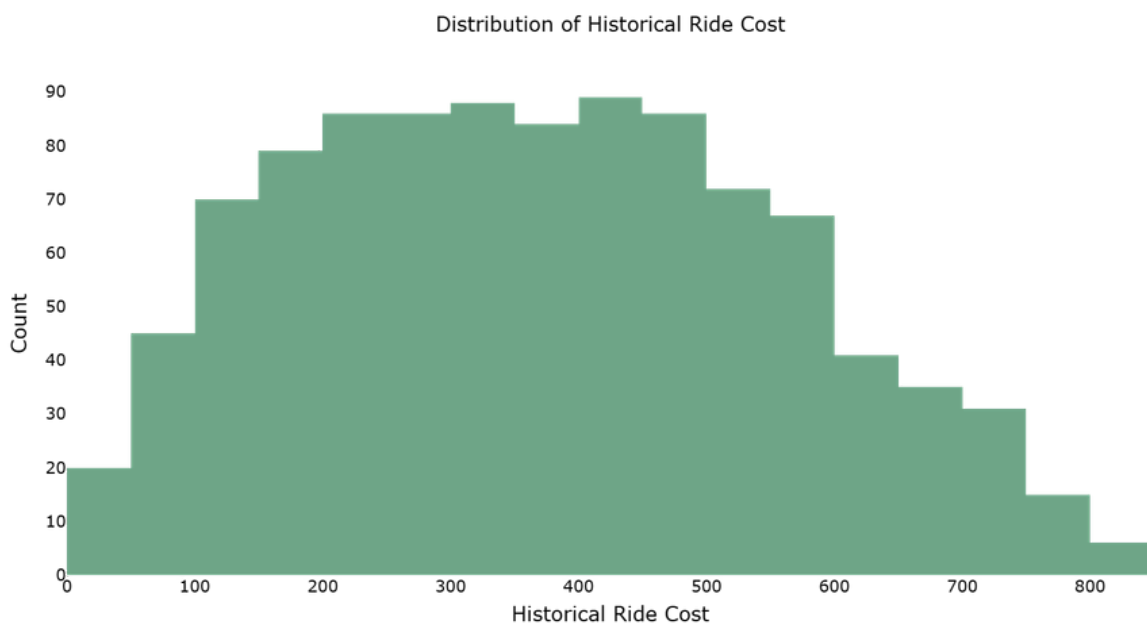
2.4 Exploratory Data Analysis



Historical ride costs are broadly distributed across a wide range, with the highest concentration occurring in the mid-range.

\$2,135

Average Cost per Service



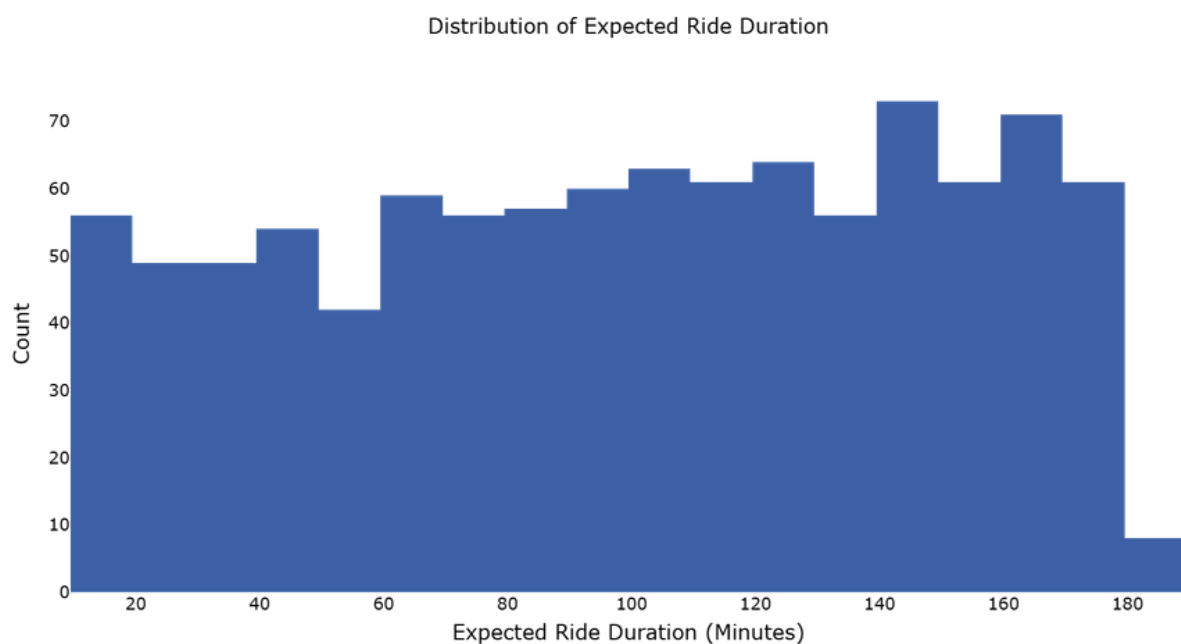
Key observations

- Historical ride costs span approximately \$25 to \$836, with the highest concentration of observations occurring roughly between \$150 and \$500.
- The distribution is reasonably broad, with observations extending across both lower-cost and higher-cost rides rather than being concentrated around a single narrow price range.

Business Insights

- The wide cost range indicates that the pricing model needs to account for multiple ride characteristics rather than relying on a single fixed fare level.
- The concentration of rides in the mid-cost range provides a substantial base of observations for learning typical pricing patterns, while higher-cost rides require the model to handle greater variation.

Expected ride durations are distributed across a broad operating range, with observations spread relatively consistently across most durations.



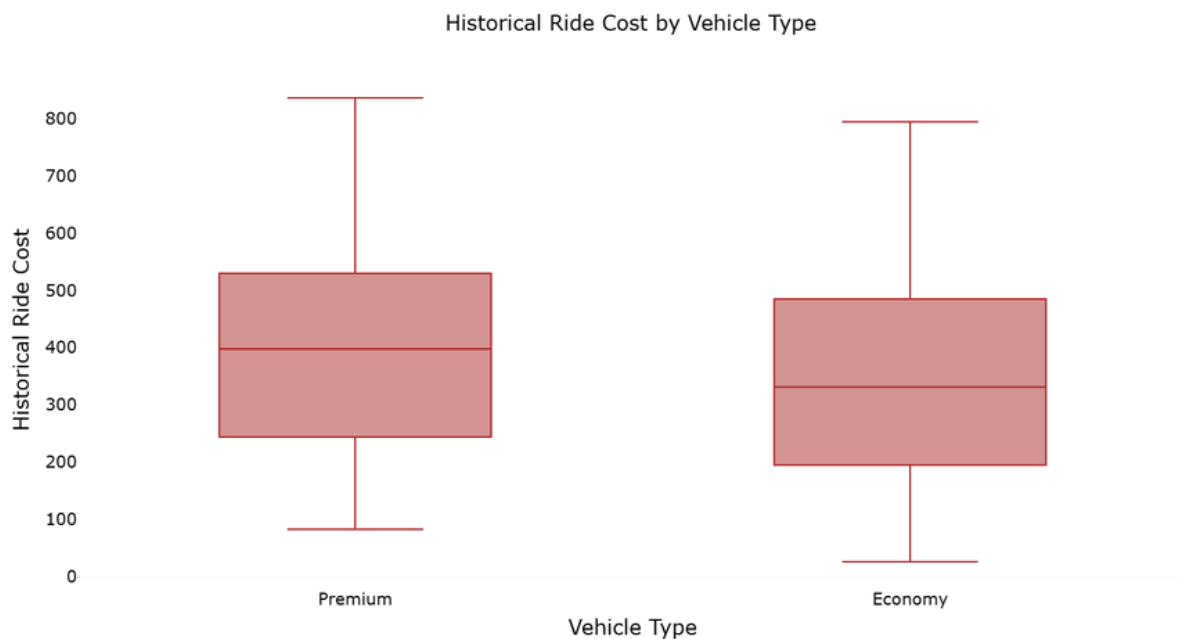
Key observations

- Expected ride duration ranges from approximately 10 to 180 minutes, with observations distributed across most of this range.
- The distribution does not show a strong concentration around one duration; several intervals between approximately 60 and 175 minutes contain relatively high observation counts.

Business Insights

- The broad duration range provides the model with observations covering both shorter and longer rides, supporting the analysis of duration-related pricing patterns.
- Since ride duration varies substantially across trips, incorporating Expected_Ride_Duration into the pricing model is important for capturing differences in historical ride costs.

Premium rides generally have higher historical costs than Economy rides, while both vehicle types show substantial cost variation.



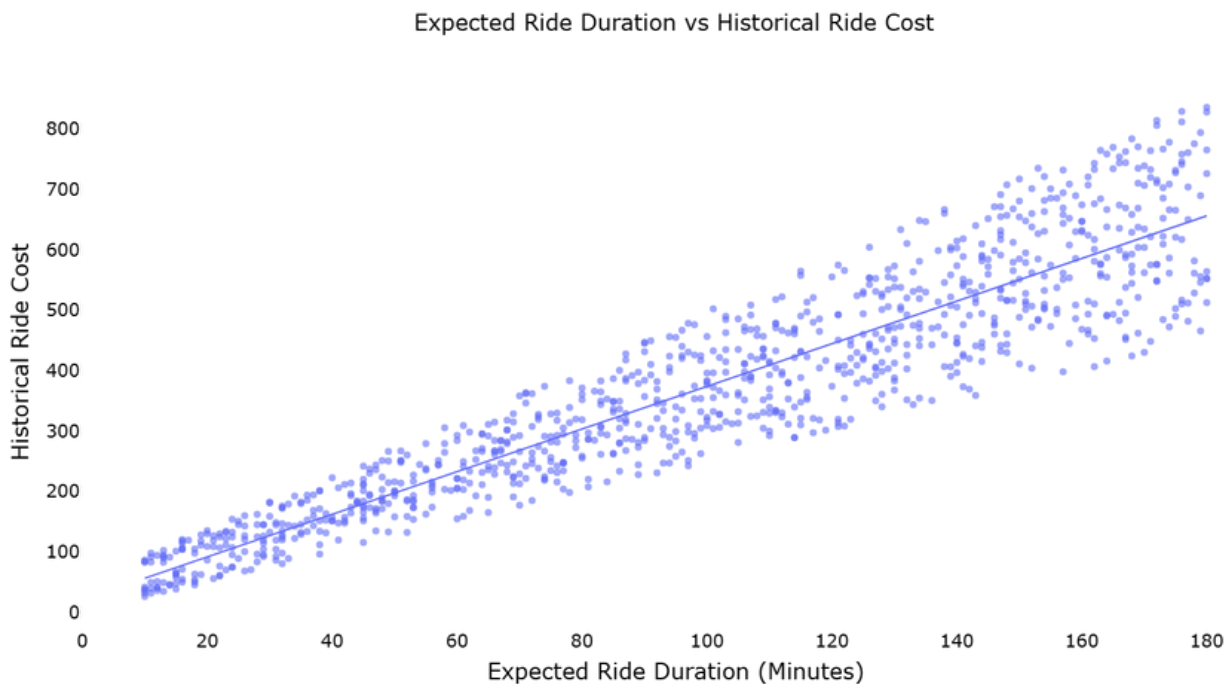
Key observations

- The median historical cost for Premium rides is visibly higher than the median for Economy rides, with the Premium median around \$400 compared with roughly \$330 for Economy.
- Both vehicle types have wide cost ranges, extending from relatively low-cost rides to values approaching or exceeding \$800, indicating that vehicle type alone does not explain the full variation in historical ride costs.

Business Insights

- Vehicle type should be incorporated into pricing models because the observed cost distributions differ between Premium and Economy rides.
- The substantial variation within each vehicle category indicates that pricing decisions should combine vehicle type with other ride characteristics rather than applying a fixed vehicle-specific price.

Historical ride cost shows a strong positive relationship with expected ride duration



Key observations

- Historical ride cost increases consistently as expected ride duration increases, with the plotted relationship showing a strong positive trend across the full duration range.
- The spread of ride costs becomes wider at longer durations, particularly beyond approximately 120 minutes, indicating greater variation in historical costs for longer rides.

Business Insights

- Expected ride duration should be an important input in ride-cost prediction because longer rides are consistently associated with higher historical costs.
- The increasing variation at longer durations indicates that additional ride characteristics are needed to explain differences in cost beyond duration alone.

Driver availability generally increases as the number of riders increases, while supply variation becomes wider at higher rider volumes



Key observations

- The scatter plot shows a clear positive relationship between the number of riders and the number of drivers, indicating that higher rider volumes are generally accompanied by greater driver availability.
- At higher rider counts, the range of observed driver counts becomes substantially wider, showing greater variation in available supply under higher demand conditions.

Business Insights

- Pricing analysis should consider both rider demand and driver supply rather than treating either variable independently.
- The variation in driver availability at higher rider volumes supports the use of a combined demand-supply feature such as Rider_Driver_Ratio to represent the balance between riders and drivers.

Historical ride costs show relatively similar distributions across booking periods, with only moderate differences in median cost.

52.03%

Typical CPU Utilization (Median)



Key observations

- The median historical cost is highest for Afternoon bookings, followed closely by Morning, while Night has the lowest median among the four periods.
- The distributions overlap substantially across all four booking periods, and each period shows a wide range of historical ride costs.

Business Insights

- Time of booking can be retained as a pricing-related feature, but the substantial overlap suggests that booking period alone does not determine historical ride cost.
- Pricing analysis should combine booking time with stronger ride-level predictors such as expected duration and vehicle type rather than applying substantially different pricing solely by time period.

The positive duration-cost relationship remains consistent across vehicle types and locations, with Premium rides generally positioned above Economy rides

51.63%

Typical Memory Utilization (Median)



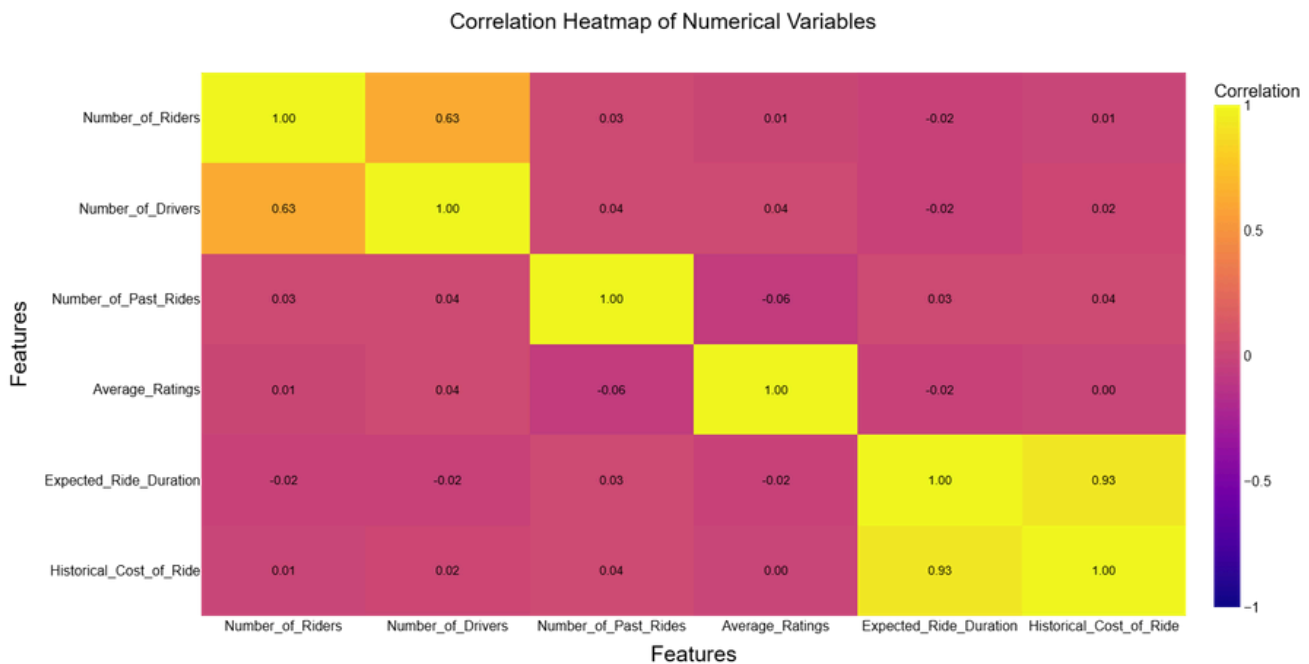
Key observations

- Historical ride cost increases with expected ride duration across Urban, Suburban, and Rural locations, showing a consistent positive relationship in all three panels.
- Within each location, Premium rides generally show higher historical costs than Economy rides at comparable ride durations, although the two groups overlap.

Business Insights

- Ride duration, vehicle type, and location should be considered together when analyzing historical ride costs rather than evaluating each factor independently.
- The consistent pattern across locations suggests that the relationship between duration and historical cost is not limited to a single geographic category.

Expected ride duration has the strongest numerical relationship with historical ride cost, while rider and driver counts show a moderate relationship with each other.



Key Observations

- Expected_Ride_Duration has a very strong positive correlation with Historical_Cost_of_Ride of approximately 0.93, making it the strongest numerical relationship with the target.
- Number_of_Riders and Number_of_Drivers have a moderate positive correlation of approximately 0.63, while the remaining numerical variables show correlations close to zero with historical ride cost.

Business Insights

- Expected ride duration should receive substantial attention in ride-cost modeling because of its strong observed relationship with historical ride cost.
- The moderate rider-driver relationship supports examining demand and supply together, which is consistent with the engineered Rider_Driver_Ratio feature used during model preparation.